# Alzheimer's Disease Progression Detection with ConvNeXt Neural Network Model

Rajcsányi Péter, Kiss Attila

***Abstract*—Alzheimer's Disease (AD) is a neurodegenerative disease, that has become more and more common in our aging society. As currently incurable, the ultimate task is to detect it preliminary, so that medical intervention can slow progression and delay certain symptoms. Several studies provided approaches for AD classification or progression detection from different biomarkers, one of the most prominent being brain atrophy, which is generally analyzed with magnetic-resonance imagery (MRI) scans. To put emphasis on progression detection in order for early intervention, this paper presents a method, which can detect progression between three brain funcionality stages: normal cognitive (NC), mild cognitive impairment (MCI) and AD, from two longitudinal MRI scans using the OASIS-2 dataset. The approach employs a Siamese neural network structure and ConvNeXt model for feature extraction. [RESULTS]**

***Index Terms*—Alzheimer's Disease, Neural Network, ConvNeXt**

## I. Introduction

Alzheimer's Disease is a progressive neurodegenerative disorder, the most common type of dementia, that may contribute to 60-70% of cases, and has become a rapidly increasing global public health concern, with millions of individuals affected. Producing symptoms of severe impact on everyday life, such as confusion, memory loss, problems with speaking, writing, etc., AD puts a heavy load on the effected individuals, their families and caretakers, global healthcare, etc. [1] [2] Recognizing the progression to Alzheimer's is crucial, due to the fact that there are treatments available only for the earlier stages of AD, including NC and MCI. [3] Unfortunately, limited awareness and difficult recognition results in approximately 67% of patients being diagnosed in the moderate stage, marking this field of research of great consequence and importance. [4] Hence, several studies are concerned with the classification of patients, or progression detection, from numerous imagining and biological markers, such as brain atrophy (or brain volume), cerebrospinal fluid, neuropsychological tests, etc. [5] [6] This method utilizes brain atrophy information, which is usually acquired using MRI, positron emission tomography (PET), or functional magnetic resonance imaging (fMRI). The proposed approach uses two MRI images, from two different scans, that had an at least 1 year long interval between them. A Siamese neural network has the two images as inputs, and provides an output of three different classifications: the patient's cognitive stage progressed, reverted, or remained the same. The feature extraction happens with the help of a model, that is based on ConvNeXt and has been modified to handle 3D data.

## II. Data

The used dataset is the OASIS-2, Cross-sectional MRLongitudinal MRI Data in Nondemented and Demented Older AdultsI Data in Young, Middle Aged, Nondemented and Demented Older Adults dataset. It consists of a longitudinal collection of 150 subjects aged 60 to 96. Each subject was scanned on two or more visits, separated by at least one year for a total of 373 imaging sessions. For each subject, 3 or 4 individual T1-weighted MRI scans obtained in single scan sessions are included. The subjects are all right-handed and include both men and women. 72 of the subjects were characterized as nondemented throughout the study. 64 of the included subjects were characterized as demented at the time of their initial visits and remained so for subsequent scans, including 51 individuals with mild to moderate Alzheimer's disease. Another 14 subjects were characterized as nondemented at the time of their initial visit and were subsequently characterized as demented at a later visit. [7] To extend the model's accuracy and classification ability, the patients's exact Clinical Dementia Rating (CDR) score was used. Fig. 1 presents a pair of MRI image segments of a progressed patient.

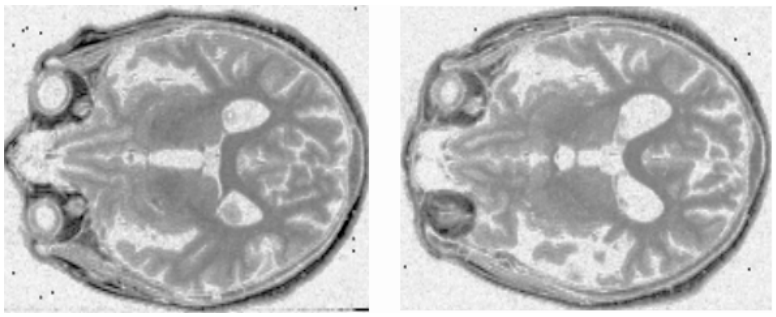


Fig. 1. MRI image segments of a patient at 88 and 92 years old. They have progressed from 0 to 0.5 CDR.

## III. Neural Network

The neural network model follows a Siamese structure, with the two MRI images as inputs. The images go into their own separate feature extractor networks, which share their weights, and are based on the ConvNeXt architecture [8]. It has been modified to handle 3D images instead of the original 2D, swapping the 2D convolutional layers for 3D ones. After feature extraction, the two outputs of the networks are connected together with a densely-connected neural network. It has 3 layers, the first two with GELU, the last one with Softmax activation function. This gives an output of 3 different classifications: 0 means the cognitive state reverted, 1 means the cognitive state stayed stable, 2 means it has progressed. Fig. 2 presents the network architecture as a whole.

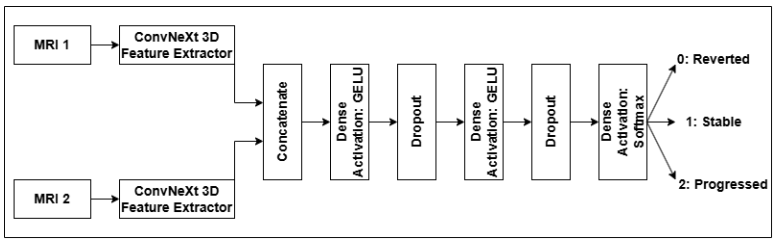



Fig. 2. Proposed neural network architecture.

## IV. Training and results

Due to resource and performance issues, the training process could not finish yet, so the project is incomplete at the time. A prior,

semifinished training with rougher parameters yielded an estimate 88-92% accuracy on training and validation. When enough resources will be available, the training will be completed and the research closed.

## V. Related work

Several researches presented high-accuracy results in classification or progression detection, using various biomarkers and methods, including but not limited to the few mentioned below.

- Yong-Shiuan Lee and Huimei Liu combined functional principal component analysis with long short-term memory (LSTM) networks to predict AD in its early stages. Data used has been obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset (adni.loni.usc.edu). They have achieved a general accuracy of 72.22% on patients with NC, MCI and dementia. [9]
- Tong Tong, Qinquan Gao, Ricardo Guerrero, Christian Ledig, Liang Chen, and Daniel Rueckert, using the ADNI dataset, has proposed a novel grading biomarker to predict patient progression from MCI to AD. The biomarker has achieved an area under the receiver operating characteristic curve (AUC) in the range of 79–81% for the prediction of MCI-to-AD conversion within three years in tenfold cross validations. The classification AUC further increases to 84–92% when age and cognitive measures are combined with the proposed grading biomarker. [10]
- Javier Escudero, John P. Zajicek, and Emmanuel Ifeachor, with the help of ADNI, proposed the creation of the Bioprofile of AD. The Bioprofile revealed key patterns of AD in the subject's biodata, that could even help the early prediction at the MCI stage, since it divided the MCI subjects into groups with different rates of conversion to AD. [11]

## VI. Conclusion

Due to resource and performance issues, the training process could not finish yet, so the project is incomplete at the time. A prior, semifinished training with rougher parameters yielded an estimate 88-92% accuracy on training and validation. When enough resources will be available, the training will be completed and the research closed.

## VII. Future works

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